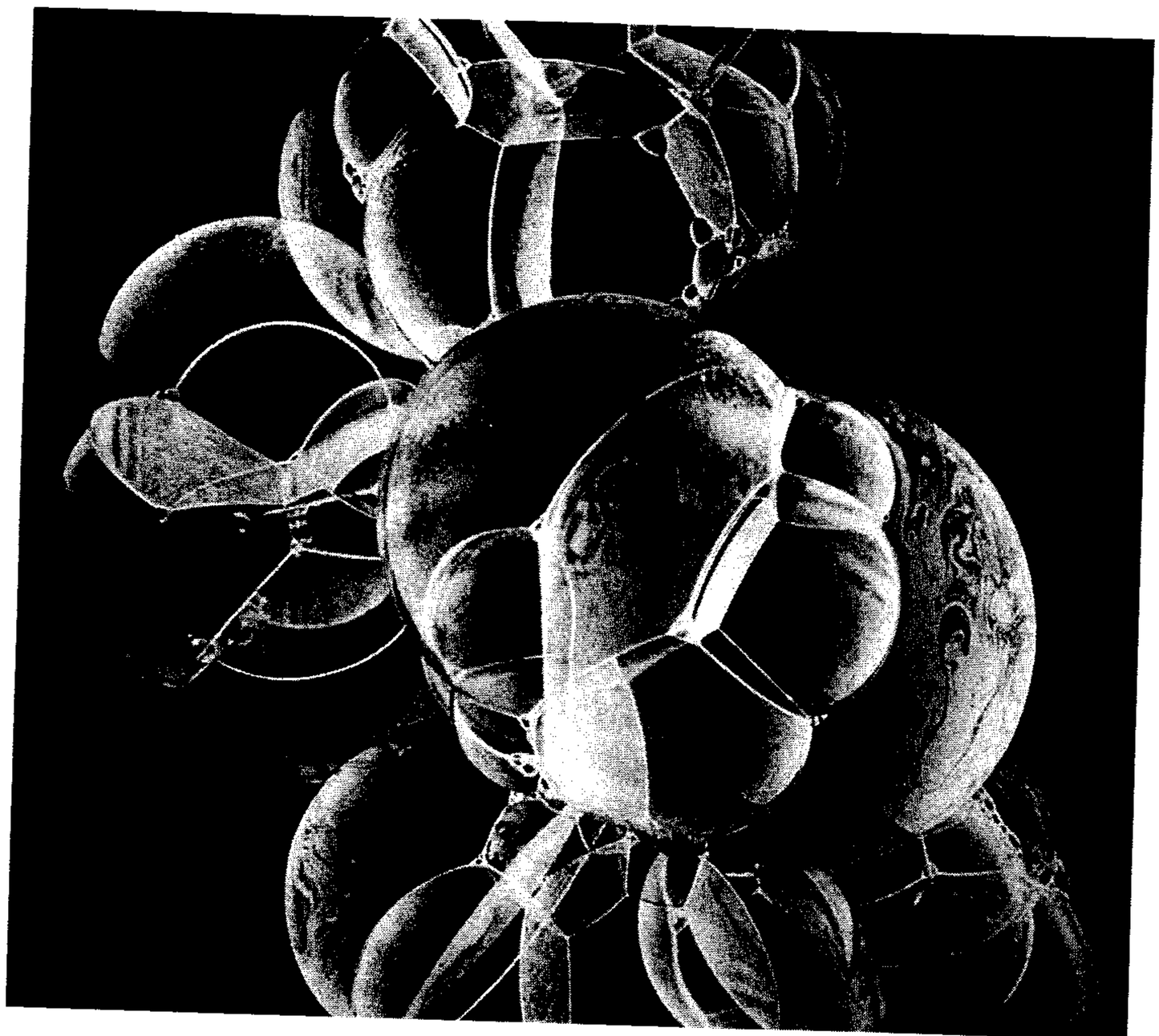
The Use of Neural Networks in Data Mining

by Welton Benjamin



Market researchers have had a long and sometimes uneasy relationship with statistical analysis. One might even venture to say that for some researchers statistical analysis is the analytical equivalent of the eccentric uncle who is not talked about in polite family conversation–yes it is a part of the market research family but please, let's just leave it at that.

Data mining cousins

Nonetheless the influence of statistics has been inescapable; techniques have been used in tasks as varied as determining sample sizes, testing for 'significant' differences between groups, and for developing sophisticated segmentation systems. Now some of these techniques, for many years used by researchers to gain useful insights into behavior, are now being joined by a new cousin: data mining.

Data mining has been increasing in popularity in recent years. While it can be used for purposes similar to survey-based research, it usually eschews the use of survey data and relies on information held in a company's data systems for exploring marketing opportunities. And one of the more popular tools used by data miners today is artificial neural networks.

What are these...neural nets?

Neural networks refer to a set of tools that are used to perform a variety of tasks that include prediction (especially for direct mail campaigns), classification, clustering and time series analysis. The data used are commonly found on databases, which are sometimes merged with demographic information from secondary sources. While researchers and statisticians have carried out these tasks for some time with other techniques, neural nets have an advantage over other procedures in one important respect: the basic assumptions that are required by some commonly used statistical techniques do not have to be fulfilled for neural nets to work well. One notable example is found in prediction. Some traditional techniques require that a unit change in an input variable relate to a similar change in the predicted variable. Neural networks have been shown to make accurate predictions where relatively complex relationships exist among variables.

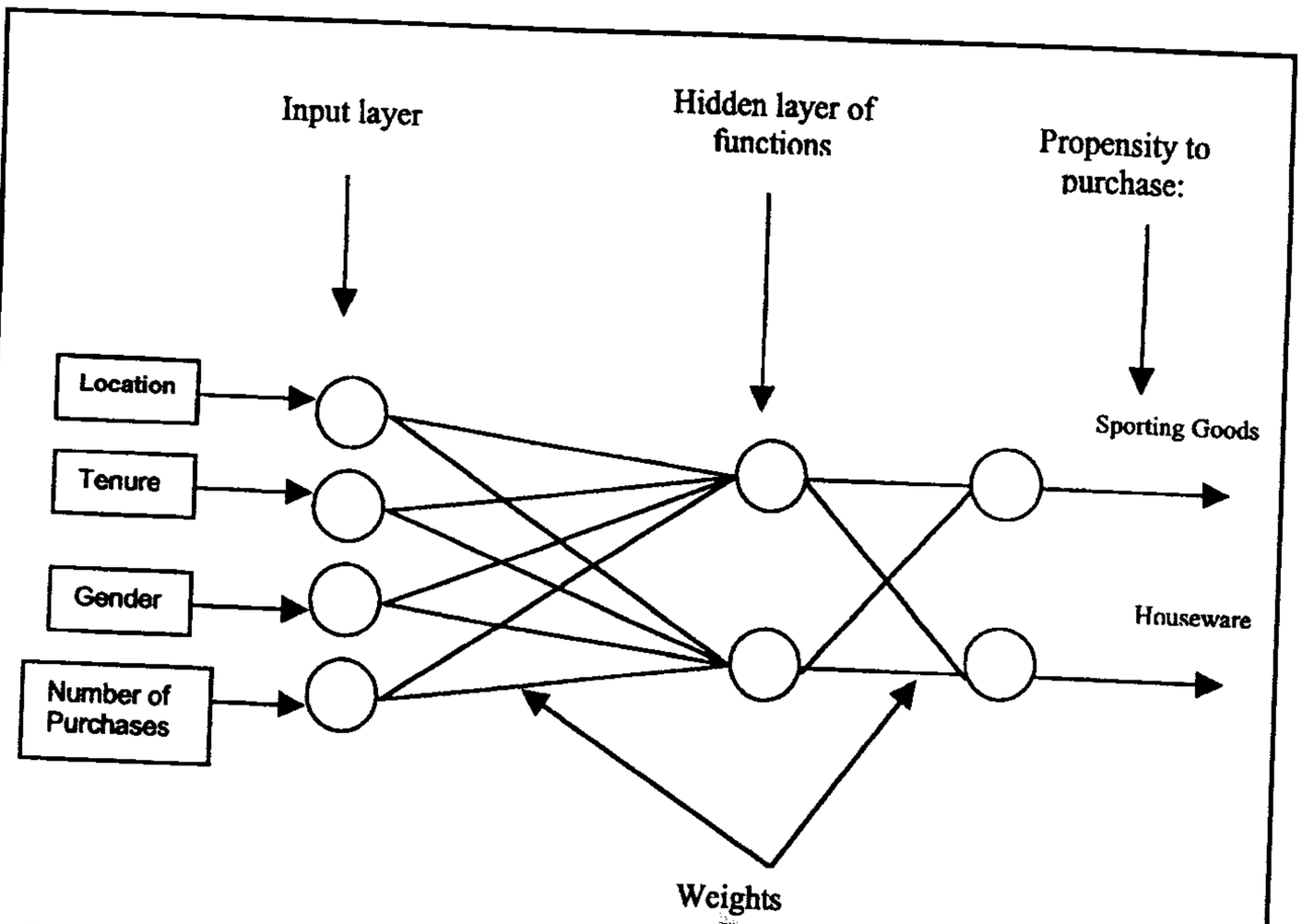
Neural networks are designed to simulate the manner in which the human brain works. The word 'neural' alludes to the millions of connections in the brain called 'neurons', which facilitate the learning process. In neural network models, mathematical functions referred to as nodes, process and pass along information to each other in parallel, uncovering complex relationships in the data quickly. It is this ability to work with complex relationships that sets neural networks apart from other techniques. The theory suggests that the networks 'learn' from data the way a person learns from experiences, and are

then able to apply that learning to new, heretofore unseen data, to perform tasks like making predictions and classifications.

How about an example?

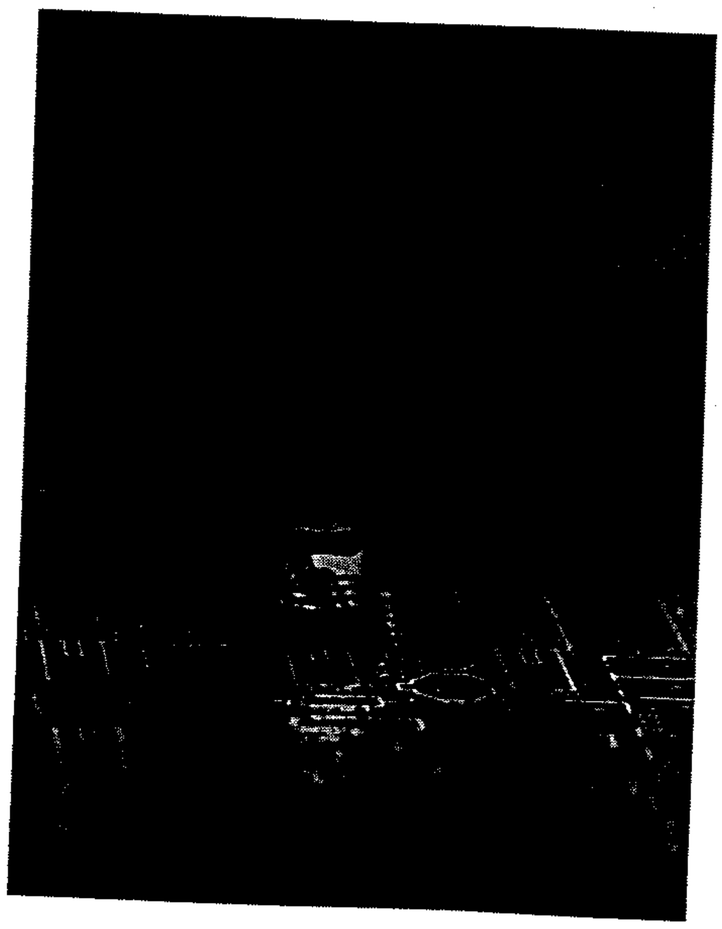
In some instances these techniques can be viewed as an alternative to more traditional marketing research methods. Consider the following scenario. A department store chain attempts to increase the amount of sales that are made on its credit cards in specific departments. To attain this goal a usage and attitude study among credit card holders might be conducted. The objectives of the study may be to determine: the products for which credit cards are most likely used, how much is spent on purchases in a specific period, what factors determine whether or not a credit card is used, the perceptions of one card vs. those of the competition, and so on. From this information a picture is painted on who uses credit cards and why, and consequently, how to increase the use of these cards. On the other hand the chain may decide to utilize the data already residing on their data systems through data mining. To do this a decision is made to identify the variables that are predictive of credit card purchases in different departments, then leverage that learning to increase sales in those areas.

The hypothetical example below illustrates how this information might be achieved. First, samples of the customer database are extracted. One sample is used for developing or 'training' the model, another for validating it. The data is then collected and cleansed--issues such as missing values and outliers may have to be dealt with. Say the following information is available: store location; the number of purchases made in different departments; tenure--the length of time the customer has held the card; and gender of the cardholder. (Other potential input variables could include household income and family size, which could be estimated from secondary sources if not available directly from the store's database.) In addition: the proportion of total credit card purchases that are made in each department is calculated. This will become the target variable, the variable that the model is to predict. The data is fed into the network and a model is developed.



Model development in a neural network.
Input variables are linked to the target variables by a series of weights and mathematical functions. Weights are adjusted until values predicted for the target variables approximate the observed values.

The diagram above illustrates the data flow from input variables to resulting predictions. To simplify the procedure there are just four input variables and two output variables The input variables are fed into the input layer of the network, a series of weights, and a middle or hidden layer of math functions represented as nodes, link the input layer with the output variables. Each input variable is connected with each mathematical function, which in turn is connected to an output variable. The resulting output will be the propensity of the average cardholder to make purchases in one of two departments, say Sporting Goods and Houseware. (In this example there are just two mathematical functions in the hidden layer. More layers as well as more functions in each layer can be used. In general, the wider the hidden layer, that is, the more functions that are used, the better the prediction rate for the current data set. A potential downside though is that the model may not perform as well on new, unfamiliar data due to possible idiosyncrasies in the development data set.)



Through a series of automatic, iterative steps the relationships between the input and output variables are established. Input and target values are compared and the weights are modified so that the predicted values of the target variable become more closely aligned to its observed values. This iterative process continues until the change in the weights become so small that they become insignificant, or a maximum number of iterations is reached. The model

that has been established is then run on the validation data to see how it performs. Depending on its level of success the model is either further adjusted to optimize performance or applied to the entire database.

If successful, this process would result in each cardholder being assigned two 'user propensities', essentially measures identifying the cardholder's likelihood of making a purchase in each of the two departments. This information could then be used to make special offerings to those cardholders most likely to respond favorably. Targeting specific customers is more likely to be more cost effective than utilizing the entire customer database. In addition, the learning from this exercise could provide insights on the buttons to push, (that is, the variables to exploit) in order to grow credit card purchases in the two departments.

Synergies may be achieved by using the data mining and customer survey in tandem. The learning obtained in the data mining phase could then be used to develop a more effective questionnaire, or to conduct the survey phase more economically. For example, information on attitude and image could be collected based on areas that were proven especially predictive on the propensity to spend.

So what else can they do?

Other types of networks are used in different situations. Networks are used for customer segmentation, where the objective of the exercise is to place customers into distinct groups according to different criteria so that they can be more efficiently targeted. The principle outlined above applies. Mathematical functions are used to identify patterns in the data, which are then used to place customers into one of a number of different groups. Again complex relationships among variables can be utilized to derive these relationships more effectively than other statistical techniques. Other areas in which neural networks have been used successfully include time series analysis, credit scoring and fraud detection.

Good news and bad news

In addition to coping effectively with data problematic for other techniques, an additional advantage of neural nets is that they are relatively easy to use. Due to the 'black box' nature of the procedure the bulk of time spent in developing the models will be in data preparation. Another positive aspect of these techniques is that the models can be updated relatively cheaply. As the sales environment changes it may become necessary to adjust the model so that it can more truly reflect the changing conditions.

On the other hand a disadvantage is that the relationships they uncover, and therefore the solutions they provide, may be difficult to interpret and therefore explain. Because the solutions presented are entirely mechanical, entirely dependent on the data used, the relationships that are uncovered, while valid, may not make sense from a marketing or operational perspective. Little insight is obtained on why certain relationships work and why others don't. As a result the techniques are most applicable in situations where correct links between input data and output variables are important, and where a strong interpretation of these relationships are less important.

In conclusion...

Neural nets and other data mining techniques are becoming more and more prevalent in industry and, while they can't duplicate many of the benefits provided by more traditional research methods, they do afford advantages that are not available elsewhere. While in some circumstances they may be considered alternatives to more familiar techniques they may also be viewed as complements to them, which when used in harmony, can provide a fuller, richer framework through which marketing information can be gathered, analyzed and utilized.

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